LECTURE 13
Linear and Nonlinear Advection

References: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Chapter 4, Section 4.1.
L. A. Barba & G. F. Forsyth, 12 steps to Navier-Stokes, steps 1, 2 and 3.
Wladimir E. Banda-Barragán

Advection and the one-way wave equation

Previously we built the finite difference formulae and checked the stability test, both on the heat equation. We now apply the same analysis to a **hyperbolic problem**.

Advection implies transport by a given velocity: a quantity is carried along by the flow, and is neither produced, destroyed nor diffused on the way.

Start from the second-order wave equation, whose operator factorises:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \qquad \Longleftrightarrow \qquad \left(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\right)\left(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\right) u = 0$$

The two factors give a wave running to the right and a wave running to the left, which is **d'Alembert's solution**:

$$u_t + c\,u_x = 0 \;\text{ and }\; u_t - c\,u_x = 0 \qquad \Longrightarrow \qquad u(x,t) = F(x-ct) + G(x+ct)$$

Each factor on its own is a **one-way wave equation**. We solve the right-running one:

$$\boxed{\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0, \qquad u(x,0) = u_0(x), \qquad c = \text{constant}}$$

Characteristics

Characteristics are specific preferred directions in the space-time or physical domain along which hyperbolic PDEs simplify into ordinary differential equations (ODEs).

Imagine we follow a point that moves along a path $x = x(t)$ and write the rate of change of u seen by that point:

$$\frac{d}{dt}\,u\big(x(t),t\big) = \frac{\partial u}{\partial t} + \frac{dx}{dt}\,\frac{\partial u}{\partial x}$$

Choosing the path so that the RHS is the equation itself:

$$\frac{dx}{dt} = c \qquad \Longrightarrow \qquad \frac{du}{dt} = \frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0$$

The paths $x = x_0 + ct$ are the **characteristics**, and u is constant on each of them.

The PDE has been reduced to an ODE.

Characteristic lines are the paths in the (x,t) plane along which information travels.

The exact solution

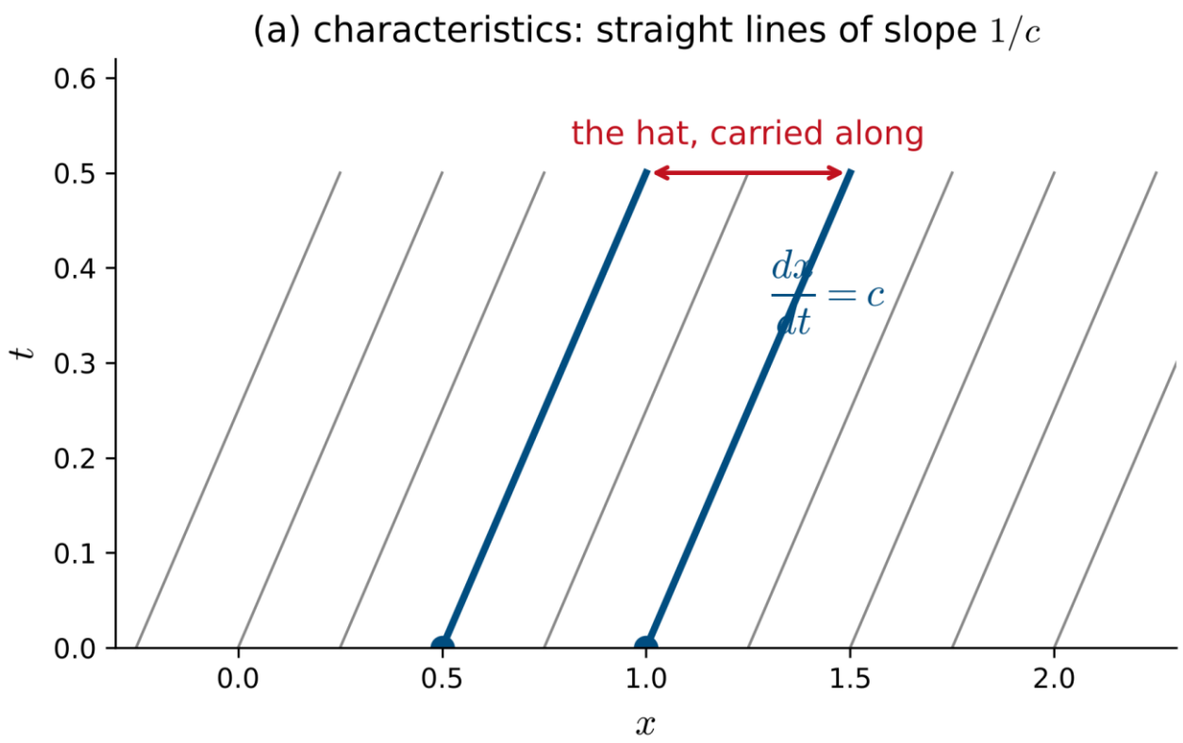
The characteristic through (x, t) starts at $x - ct$, so u carries the initial value from that point:

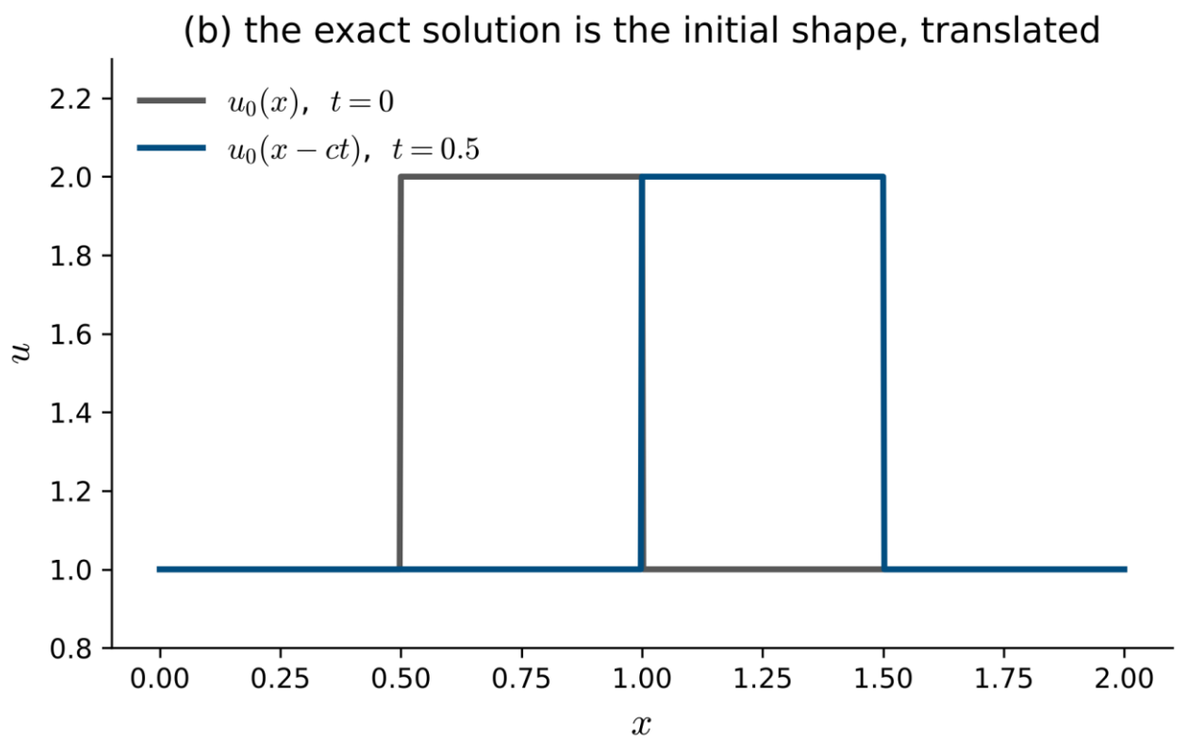
$$u(x,t) = u_0(x - ct)$$

Substituting, with the chain rule and $s = x - ct$:

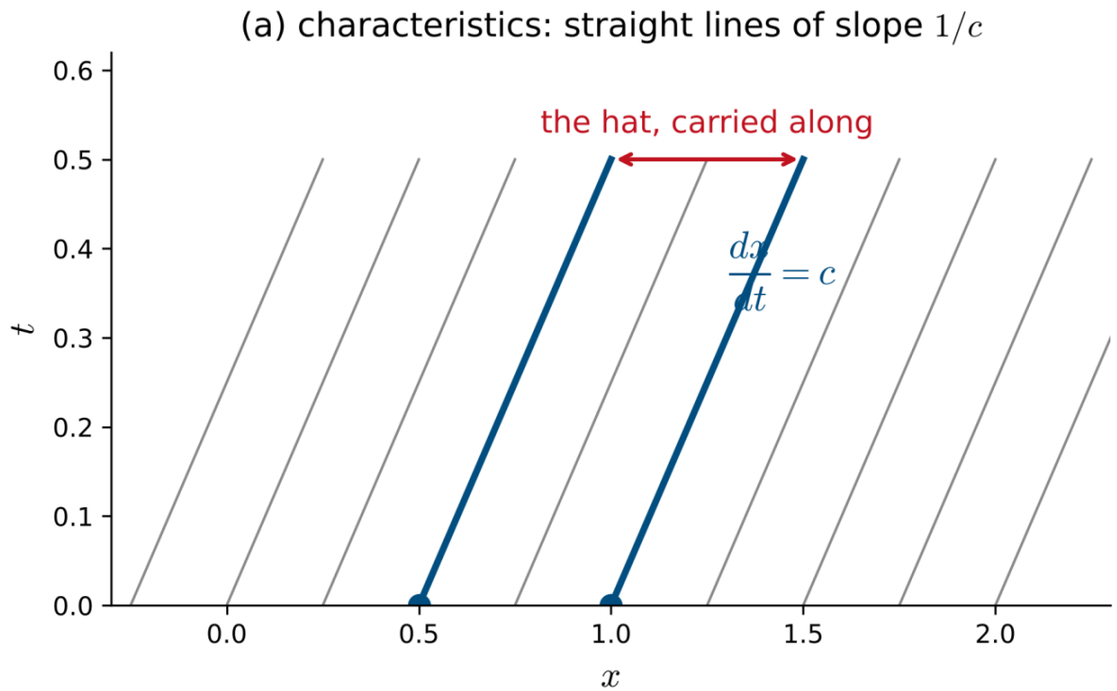
$$\frac{\partial u}{\partial t} = -c\,u_0'(s), \qquad \frac{\partial u}{\partial x} = u_0'(s) \qquad \Longrightarrow \qquad u_t + c\,u_x = -c\,u_0'(s) + c\,u_0'(s) = 0$$

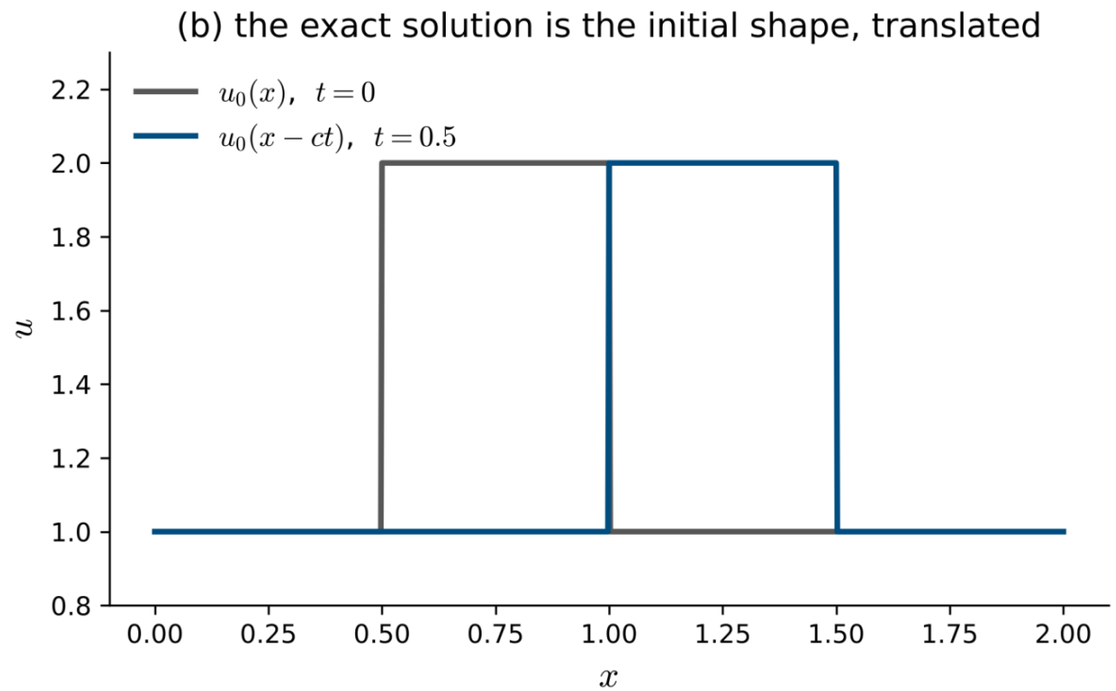
The initial profile is translated at speed c. Its amplitude and its shape do not change.





The solution in the (x, t) plane





To find u at a point, follow its characteristic back to $t = 0$.

Notation convention:

$$x \to i\,, \qquad t \to n \qquad \Longrightarrow \qquad u(x,t) \longrightarrow u_i^n\,, \quad \text{stored as } \texttt{u[i, n]}$$

$$x_i = i\,\Delta x\,, \quad i = 0, 1, \ldots, n_i - 1; \qquad t^n = n\,\Delta t\,, \quad n = 0, 1, \ldots, n_t - 1$$

Forward difference in time (LHS)

$$\boxed{\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0, \qquad u(x,0) = u_0(x), \qquad c = \text{constant}}$$

Expand about the point (i, n) at fixed x:

$$u_i^{n+1} = u_i^n + \Delta t \left(\frac{\partial u}{\partial t}\right)_i^n + \frac{(\Delta t)^2}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n + \mathcal{O}[(\Delta t)^3]$$

Solving for the first derivative gives the forward difference and its leading error term:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = \left(\frac{\partial u}{\partial t}\right)_i^n + \frac{\Delta t}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n + \mathcal{O}[(\Delta t)^2]$$

This is first order in time.

Backward difference in space (RHS)

$$\boxed{\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0, \qquad u(x,0) = u_0(x), \qquad c = \text{constant}}$$

Expand backwards in x at fixed t:

$$u_{i-1}^n = u_i^n - \Delta x \left(\frac{\partial u}{\partial x}\right)_i^n + \frac{(\Delta x)^2}{2}\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n - \mathcal{O}[(\Delta x)^3]$$

$$\frac{u_i^n - u_{i-1}^n}{\Delta x} = \left(\frac{\partial u}{\partial x}\right)_i^n - \frac{\Delta x}{2}\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + \mathcal{O}[(\Delta x)^2]$$

First order in space, with a second space derivative in the error.

The backward side is chosen because for $c > 0$ the information arrives from the left.

The finite difference equation and the Courant number

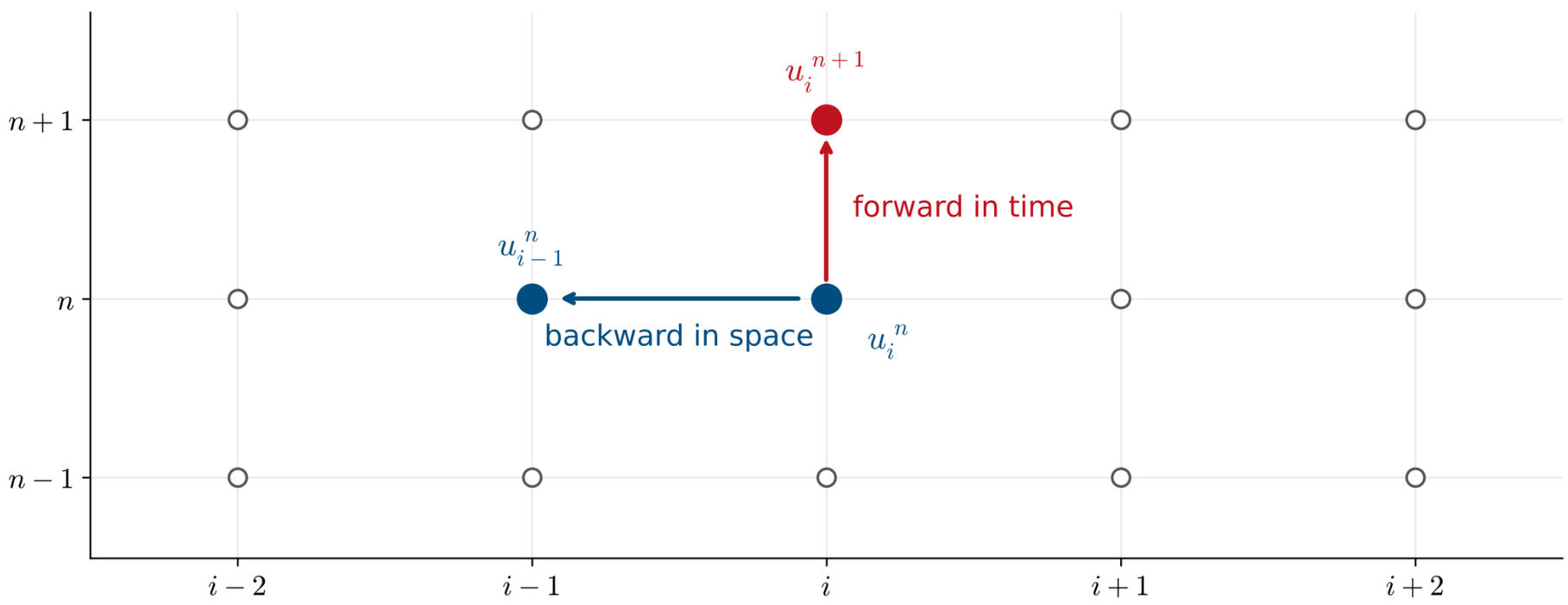
Substituting both differences into the one-way wave equation:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + c\,\frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$$

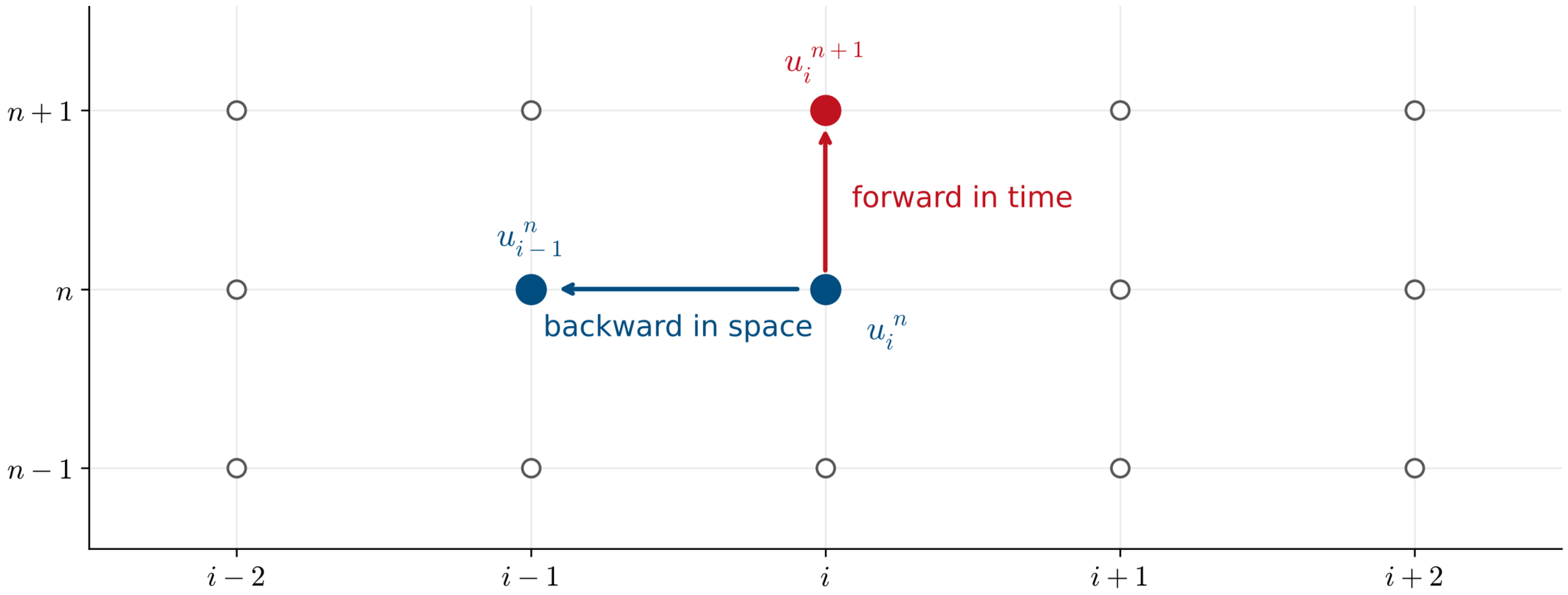
There is one unknown, so the scheme is explicit:

$$u_i^{n+1} = u_i^n - \frac{c\,\Delta t}{\Delta x}\left(u_i^n - u_{i-1}^n\right) = u_i^n - C\left(u_i^n - u_{i-1}^n\right), \qquad C \equiv \frac{c\,\Delta t}{\Delta x}$$

C is the Courant number (CFL), the number of cells the wave crosses in one time step.



The upwind stencil



Collecting the two terms on the right, the update can be seen as a weighted average:

$$u_i^{n+1} = (1 - C)\, u_i^n + C\, u_{i-1}^n$$

For $0 \leq C \leq 1$, both weights are non-negative and sum to 1, so the new value lies between its two neighbours and no new maximum or minimum can appear.

At $C = 1$ the formula reduces to $u(i, n+1) = u(i-1, n)$, which means the profile is shifted one cell per step, exactly.

Truncation error

Substituting the 2 Taylor expansions back into the FDE and keeping the error terms:

$$\underbrace{\frac{u_i^{n+1} - u_i^n}{\Delta t} + c\,\frac{u_i^n - u_{i-1}^n}{\Delta x}}_{\text{the scheme}} = \underbrace{(u_t + c\,u_x)}_{=0 \text{ for the exact } u} + \frac{\Delta t}{2}\,u_{tt} - \frac{c\,\Delta x}{2}\,u_{xx} + \mathcal{O}[(\Delta t)^2, (\Delta x)^2]$$

Both error terms vanish as the mesh is refined, so the scheme is consistent, and it is first order in Δt and in Δx.

On solutions of the equation, the time derivative can be replaced by a space derivative:

$$u_t = -c\,u_x \qquad \Longrightarrow \qquad u_{tt} = -c\,u_{xt} = -c\,(u_t)_x = c^2 u_{xx}$$

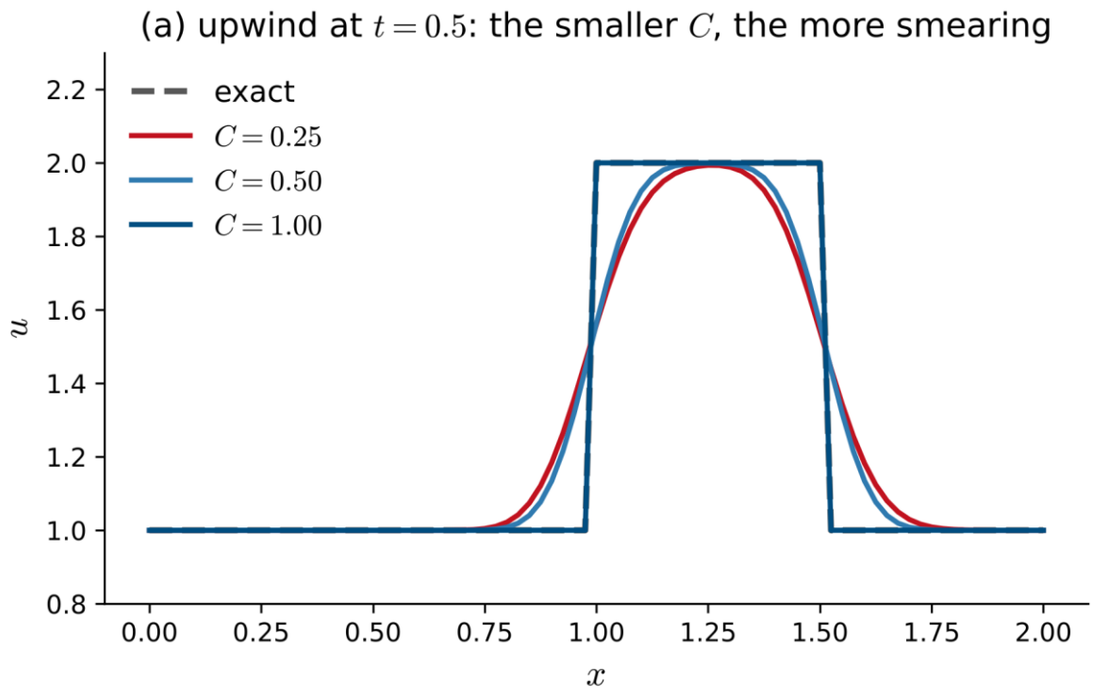
Substituting and collecting the two curvature terms:

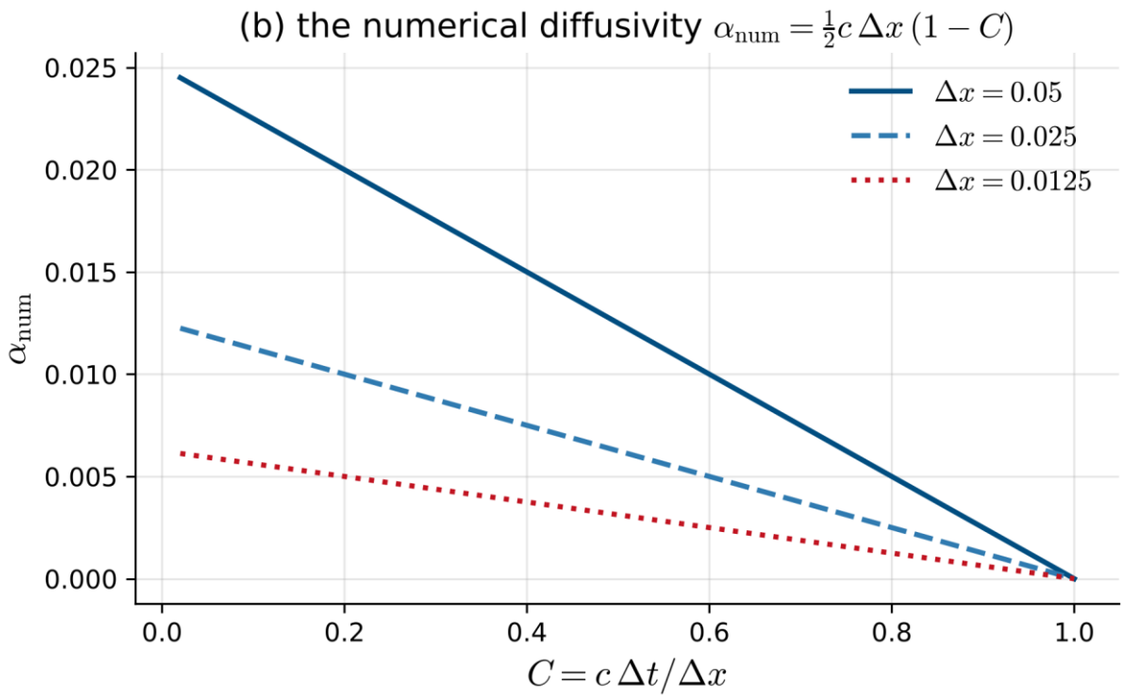
$$u_t + c\,u_x = \underbrace{\frac{c\,\Delta x}{2}\,(1 - C)}_{\alpha_{\text{num}}}\,u_{xx} + \mathcal{O}[(\Delta x)^2]$$

The scheme solves an **advection-diffusion equation**.

The numerical diffusivity is first order in Δx, positive for $0 < C < 1$, and zero at $C = 1$.

Numerical diffusion





At $C = 1$ the update is a one-cell shift and the profile is reproduced exactly.

- The plot on the right quantifies numerical diffusion: Numerical diffusivity decreases linearly as the Courant number (C) increases, reaching zero at $C = 1.0$.
- Numerical diffusivity decreases as the grid is refined: smaller grid sizes (Δx) produce lower numerical diffusivity across all CFL numbers.
- At $C = 1.0$, $\alpha_{\text{num}} = 0$, there is zero artificial smearing.

von Neumann analysis of the upwind scheme

Substitute a single Fourier mode into the difference equation:

$$u_i^n = G^n e^{\mathrm{i}\,k\,i\,\Delta x}, \qquad \theta \equiv k\,\Delta x$$

$$G = 1 - C\left(1 - e^{-\mathrm{i}\theta}\right) = \underbrace{\left[\,1 - C(1-\cos\theta)\,\right]}_{\Re G} - \mathrm{i}\,\underbrace{C\sin\theta}_{-\Im G}$$

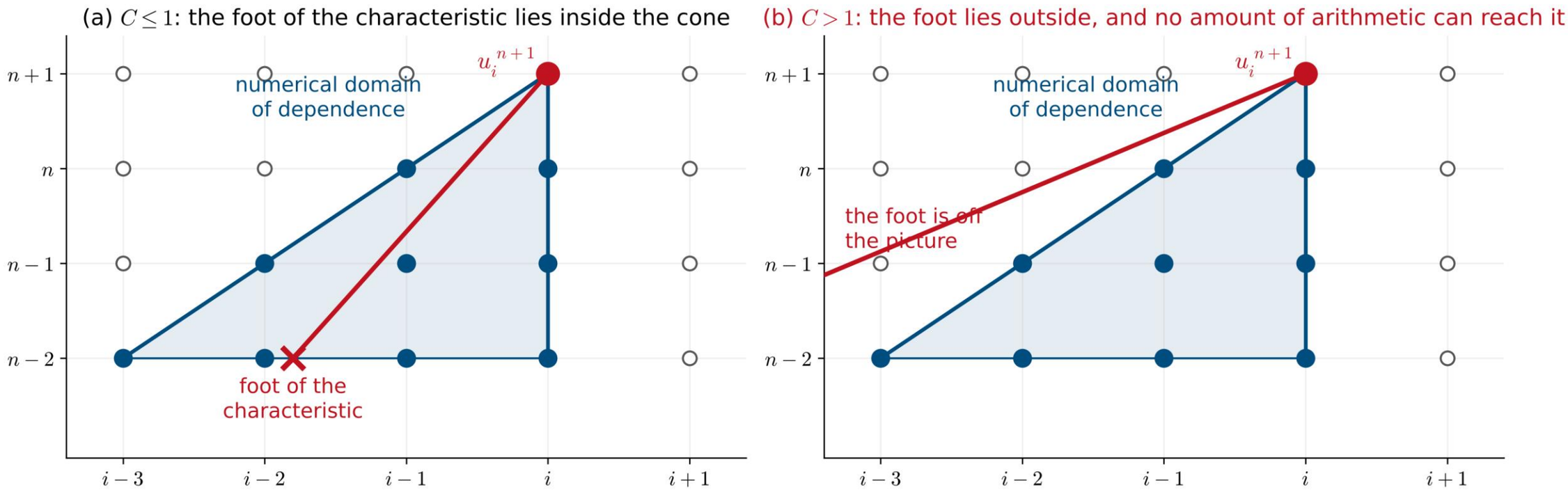
The scheme is stable when $|G| \le 1$ for every θ in $[0, \pi]$.

$$\begin{aligned} |G|^2 &= \left[1 - C(1-\cos\theta)\right]^2 + C^2\sin^2\theta \\ &= 1 - 2C(1-\cos\theta) + C^2\left[(1-\cos\theta)^2 + \sin^2\theta\right] \\ &= 1 - 2C(1-\cos\theta) + 2C^2(1-\cos\theta) \\ &= 1 - 2\,C\,(1-C)\,(1-\cos\theta) \end{aligned}$$

Since $1 - \cos\theta \ge 0$ for every mode, $|G| \le 1$ requires $C(1 - C) \ge 0$:

$$\boxed{0 \;\le\; C = \frac{c\,\Delta t}{\Delta x} \;\le\; 1 \qquad \Longleftrightarrow \qquad \Delta t \le \frac{\Delta x}{c}}$$

The CFL condition



Courant, Friedrichs and Lewy, 1928: the **numerical domain of dependence** of a point must contain its **physical domain of dependence**.

The upwind stencil reaches one cell further to the left at each step, so after n steps it covers n cells. The characteristic through the same point reaches back $c\, n\, \Delta t$. The first contains the second when $c\, \Delta t \;\leq\; \Delta x$.

If it does not, the computed value depends on data the scheme never read, and refining Δt alone does not fix it.

The condition is necessary for any explicit scheme on this stencil, and it is not sufficient: **FTCS satisfies it and is still unstable**.

Centred difference in space: FTCS

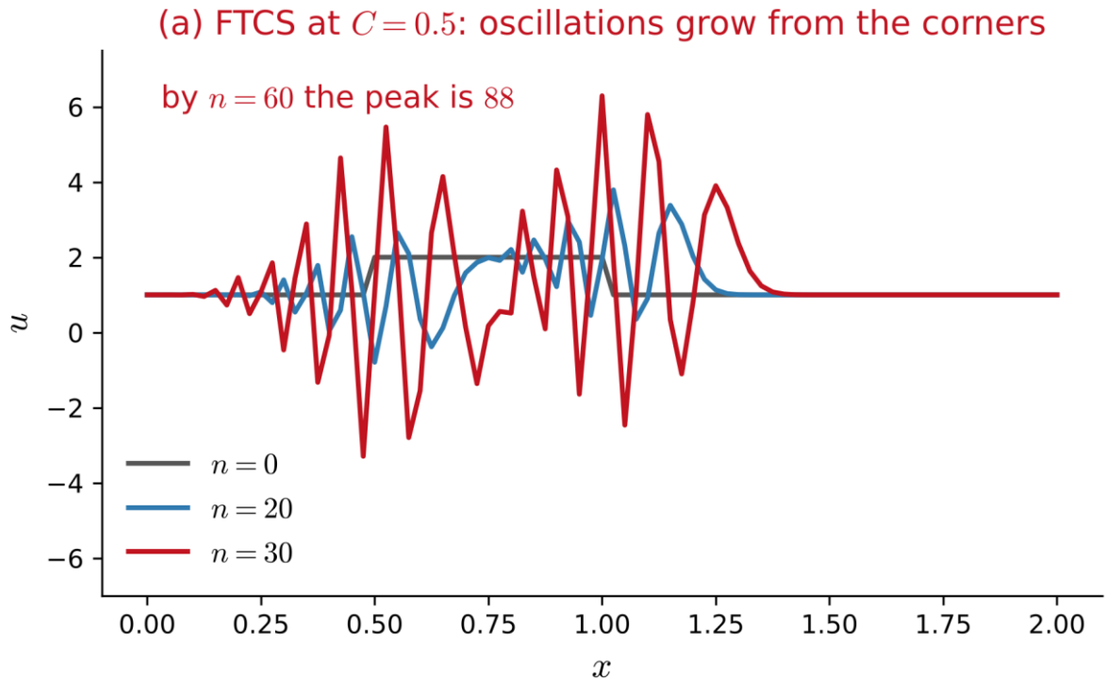
The centred difference is second order in Δx, so it is the natural alternative:

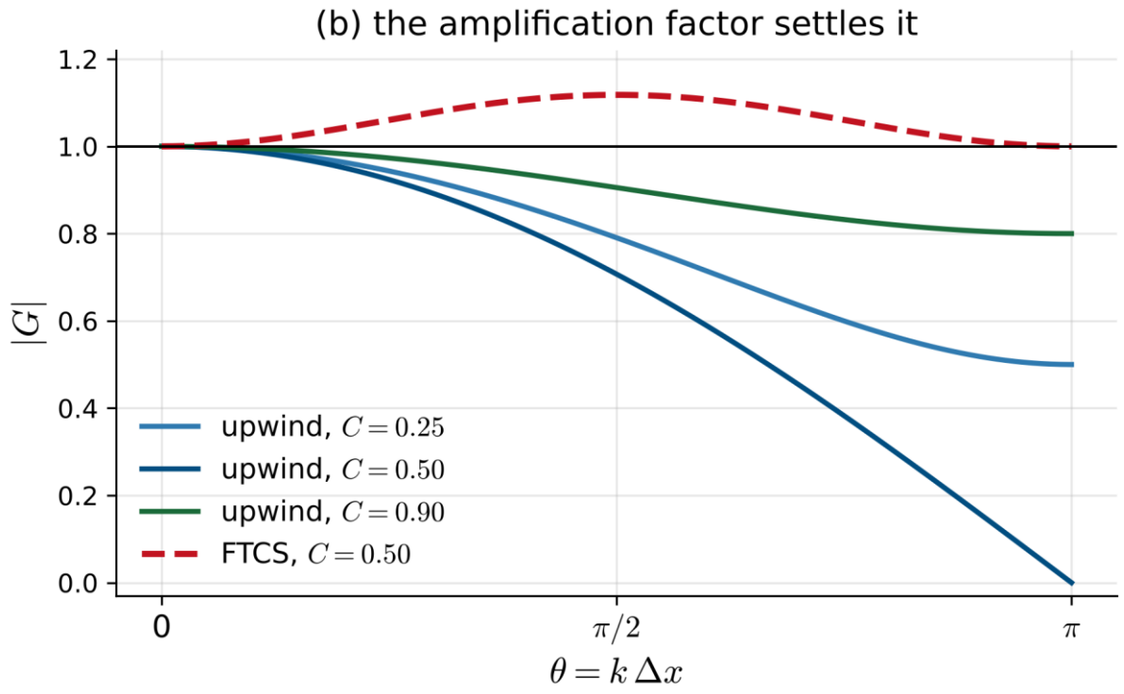
$$u_i^{n+1} = u_i^n - \frac{C}{2}\left(u_{i+1}^n - u_{i-1}^n\right) \qquad \Longrightarrow \qquad G = 1 - \frac{C}{2}\left(e^{\mathrm{i}\theta} - e^{-\mathrm{i}\theta}\right) = 1 - \mathrm{i}\, C \sin\theta$$

$$|G|^2 = 1 + C^2 \sin^2\theta \;\; > \;\; 1 \qquad \text{for every } C \neq 0 \text{ and } \sin\theta \neq 0$$

FTCS is unconditionally unstable for this equation.

No time step works: order of accuracy and stability are separate properties.





The growth begins at the shortest wavelengths, where $|G|$ is largest.

Upwinding for either sign of c

The backward difference is the correct one only when the wave arrives from the left.

In general, the stencil takes the neighbour on the upwind side:

$$u_i^{n+1} = u_i^n - \Delta t \times \begin{cases} c\,\dfrac{u_i^n - u_{i-1}^n}{\Delta x} & c > 0 \quad \text{(backward difference)} \\ c\,\dfrac{u_{i+1}^n - u_i^n}{\Delta x} & c < 0 \quad \text{(forward difference)} \end{cases}$$

The stability condition becomes $|c|\,\Delta t \,/\, \Delta x \;\leq\; 1$.

When the method is implemented in code, a conditional is needed to correctly capture the flow.

The nonlinear advection equation

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \qquad \Rightarrow \qquad \boxed{\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0}$$

The constant c has been replaced by the solution itself, u.

This is the **inviscid Burgers equation**, written in non-conservative form.

The equation is still first order and still hyperbolic.

It is no longer linear, so solutions cannot be superposed and the Fourier analysis of Lecture 12 applies only to a frozen coefficient.

Converging characteristics steepen the profile, diverging ones flatten it.

Characteristics with a variable speed

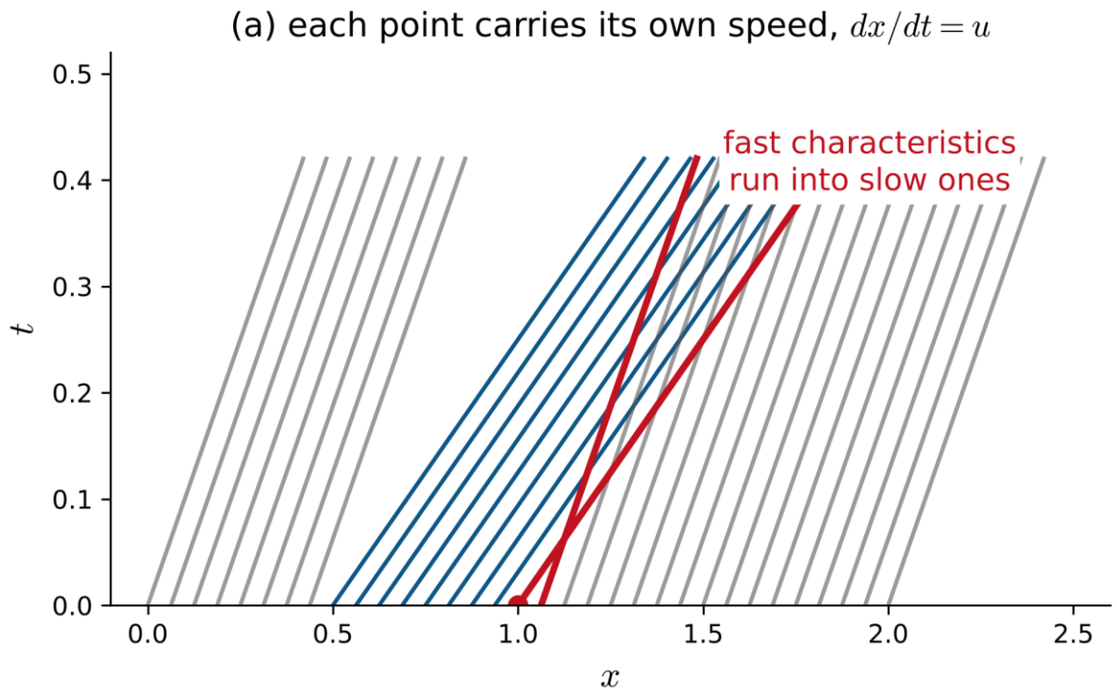
Repeating the argument, with the path moving at the local value of u:

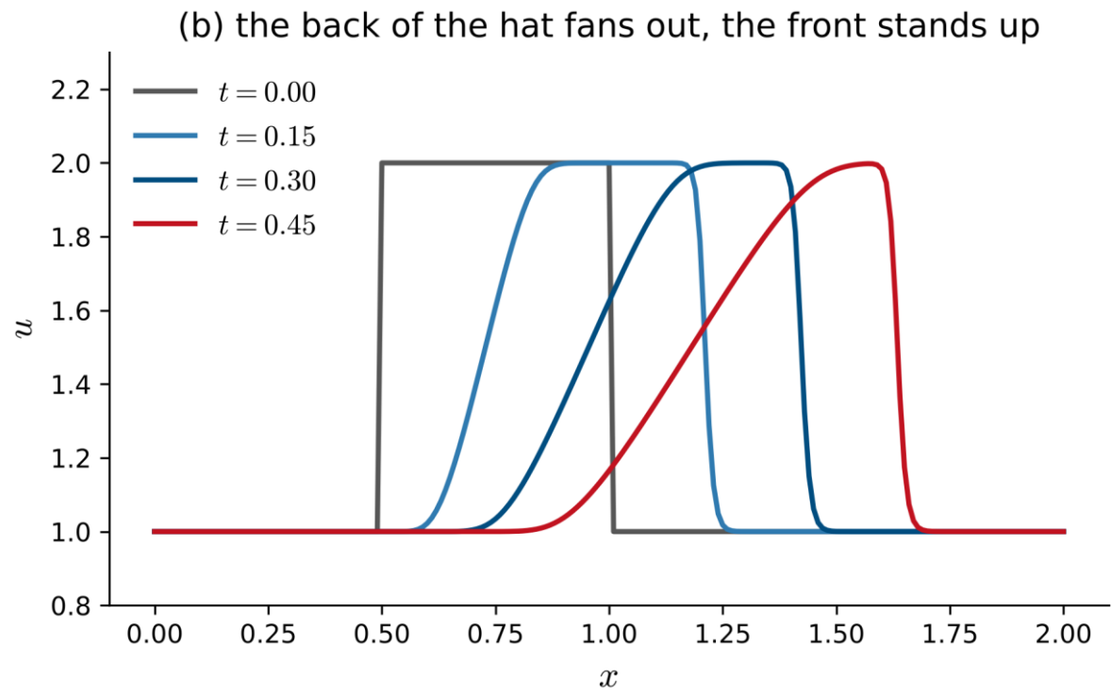
$$\frac{dx}{dt} = u \qquad \Longrightarrow \qquad \frac{du}{dt} = u_t + u\,u_x = 0$$

u is constant along its characteristic, so the characteristics are still straight lines, each with its own slope set by the initial value it carries:

$$x = \xi + u_0(\xi)\,t\,, \qquad u(x,t) = u_0(\xi) \qquad \Longrightarrow \qquad u = u_0(x - u\,t)$$

The solution is now given implicitly, because the lines depend on it.





Steepening and the breaking time

Differentiating the characteristic map with respect to its starting point, then applying the chain rule:

$$\frac{\partial x}{\partial \xi} = 1 + u_0'(\xi)\, t \qquad \Longrightarrow \qquad u_x = \frac{du_0}{d\xi}\,\frac{\partial \xi}{\partial x} = \frac{u_0'(\xi)}{1 + u_0'(\xi)\, t}$$

The slope is infinite when the denominator vanishes, which first happens at:

$$t_b = -\,\frac{1}{\min\limits_{\xi} u_0'(\xi)}, \qquad \text{finite only if } u_0' < 0 \text{ somewhere}$$

After that time, the classical solution no longer exists, although the difference scheme continues to produce numbers.

- If $u_0{}'(\xi) \geq 0$ everywhere (strictly non-decreasing profile), the wave only spreads out (rarefaction) and never breaks ($t_b \to \infty$).

- If $u_0{}'(\xi) < 0$ somewhere, the negative slope causes characteristics to cross, and the explicit negative sign ensures t_b yields a positive time value, so it breaks.

The discrete form

The same FTBS pair, with c replaced by the value at the point:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0 \qquad \Longrightarrow \qquad u_i^{n+1} = u_i^n - u_i^n \frac{\Delta t}{\Delta x}\left(u_i^n - u_{i-1}^n\right)$$

$$C_i^n = \frac{u_i^n \, \Delta t}{\Delta x}$$ varies with position and time level

Stability of the nonlinear scheme

Freezing the coefficient and repeating the Fourier argument gives the same bound, applied to the largest speed on the mesh:

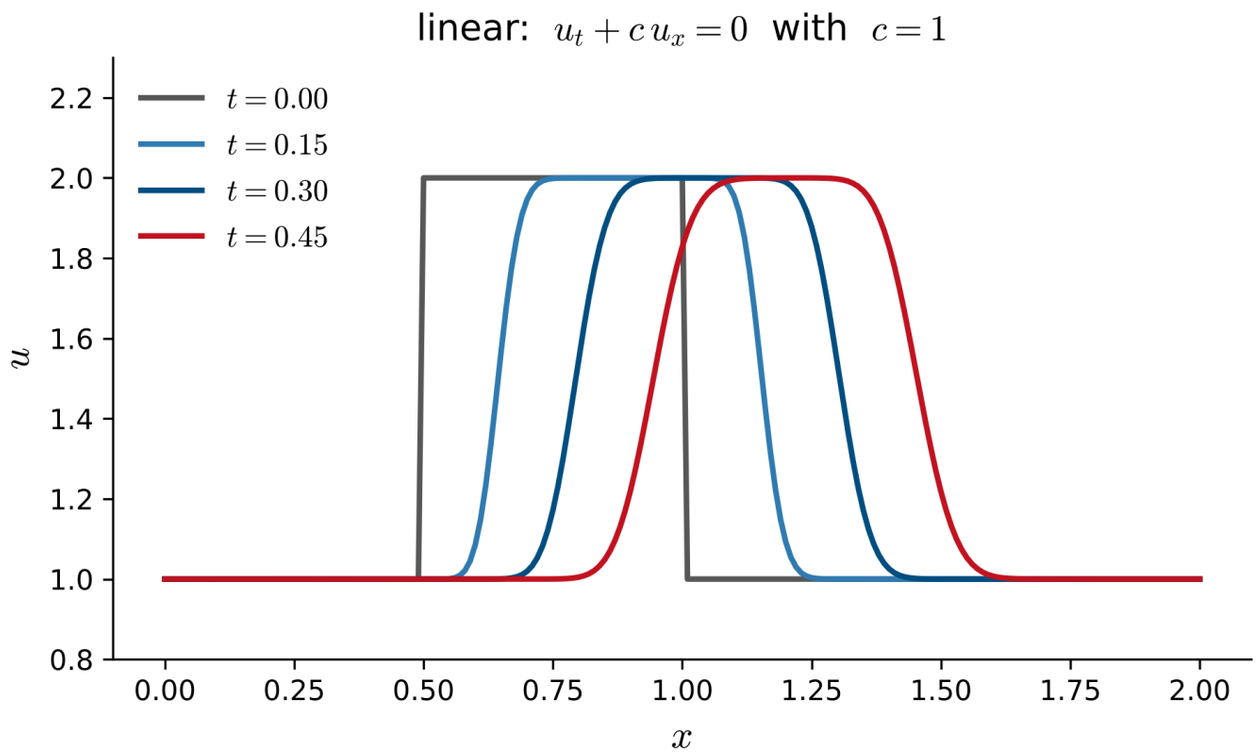
$$\Delta t \;\leq\; \frac{\Delta x}{\max_i |u_i^n|}$$

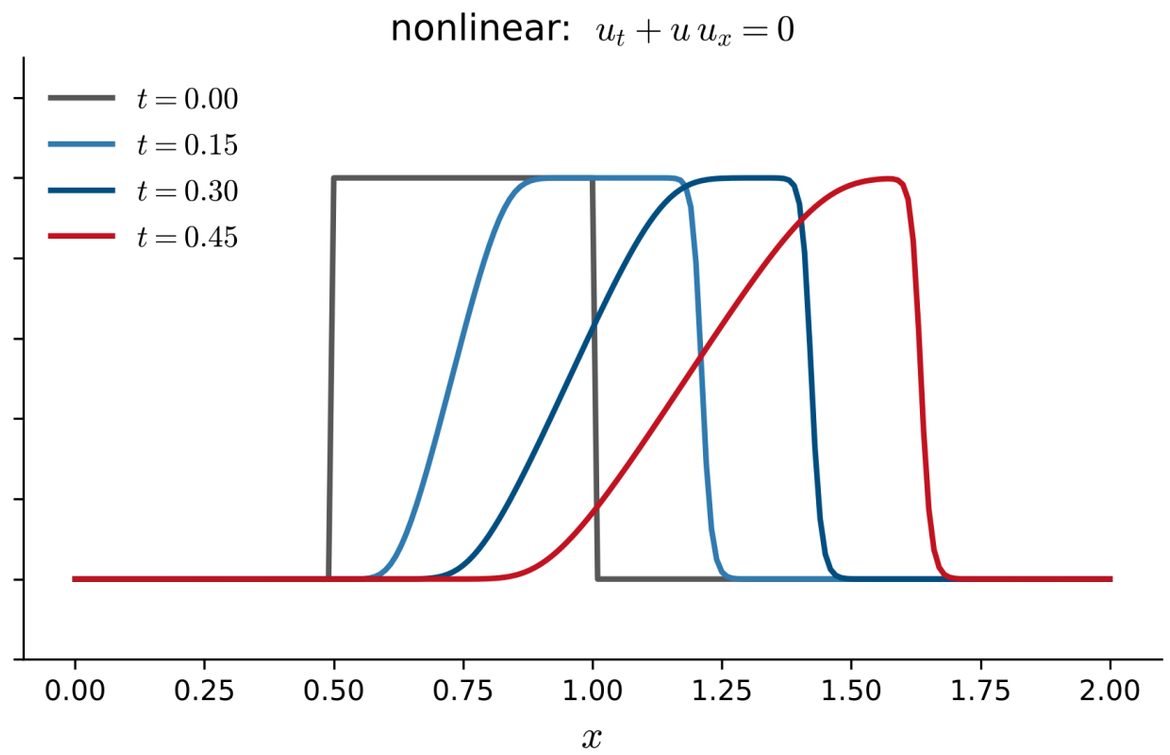
The time step therefore has to be recomputed at every step.

Compressible codes use the same rule with $\max(|u| + a)$, where a is the speed of sound.

Linear and nonlinear top-hat advection

Comparing linear and non-linear advection of a top-hat (square) signal:





Same mesh and same initial profile.

Only the speed in the convective term differs.

See notebook: https://github.com/wbandabarragan/computational-fluid-dynamics/blob/main/notebooks/nb07_linear_nonlinear_advection.ipynb

Non-conservative and conservative form

The form differenced above is non-conservative:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$$

It uses $u\, u_x$ rather than the derivative of a flux.

$$u_t + u\, u_x = 0 \qquad \text{versus} \qquad u_t + \frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) = 0$$

The two are equivalent for smooth solutions.

Once the profile breaks, they are not.

The non-conservative form propagates the discontinuity at the wrong speed. We will demonstrate this in a later lecture.

Summary

1. The one-way wave equation is derived from the second-order wave equation. Its solution is the initial profile translated at speed c (**advection process**).

2. FTBS gives an explicit update which is a weighted average of two neighbours for $0 \leq C \leq 1$.

3. The same bound follows from the **von Neumann analysis** and from the CFL argument about **domains of dependence**.

4. The modified equation gives a numerical diffusivity $\frac{1}{2} c \Delta x (1 - C)$, which is zero at $C = 1$.

5. Replacing c by u leaves the method unchanged and makes the problem **nonlinear**:

The profile steepens, the time step has to be recomputed every step, and a discontinuity forms at a finite time.